

Detecting Abnormal Trading Behavior in the Nepal Stock Exchange (NEPSE)

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Preliminary Research Note

Introduction

Financial markets are expected to incorporate publicly available information into asset prices. However, in emerging markets such as the Nepal Stock Exchange (NEPSE), trading behavior may sometimes reflect information asymmetry, speculative trading, or coordinated trading activity. Detecting abnormal trading patterns is therefore important for understanding market efficiency and identifying potential instances of insider trading or price manipulation.

This study investigates abnormal trading behavior in NEPSE by identifying unusual price and volume movements and analyzing how stock prices behave before and after these events. Using an event-study framework, the analysis examines whether abnormal trading events are preceded by systematic price movements that may suggest informed trading or information leakage.

Data

The dataset used in this study was constructed using historical daily trading data collected from ShareSansar. The dataset includes daily stock prices, trading volumes, and transaction values for NEPSE-listed companies. In addition, the NEPSE market index was collected to compute market returns and benchmark overall market performance.

The cleaned dataset contains:

Table 1. Dataset Summary

Metric	Value
Total observations	532,274
Unique stocks	353
Suspicious trading events detected	843
Study period	~11 years

These observations provide a comprehensive panel dataset of daily trading activity across the Nepal Stock Exchange.

Methodology

The analysis follows a market-adjusted event study framework commonly used in empirical finance.

Stock Returns

Daily stock returns are computed using logarithmic returns:

$$R_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right)$$

where $(P_{i,t})$ represents the closing price of stock (i) on day (t).

Market Returns

Market returns are calculated using the NEPSE index:

$$R_{m,t} = \ln \left(\frac{I_t}{I_{t-1}} \right)$$

where (I_t) represents the NEPSE index level on day (t).

Abnormal Returns

Abnormal returns measure stock performance relative to overall market movements:

$$AR_{i,t} = R_{i,t} - R_{m,t}$$

Identification of Abnormal Trading Events

Suspicious trading events are identified using two conditions:

1. Return Spike

$$|AR_{i,t}| > 3 \times \sigma_{i,30}$$

where $(\sigma_{i,30})$ represents the 30-day rolling standard deviation of stock returns.

2. Volume Spike

$$Volume_{i,t} > 5 \times AvgVolume_{i,30}$$

where the average trading volume is computed over the previous 30 trading days.

Days satisfying both conditions are classified as **abnormal trading events**.

Using these criteria, the study identified **843 abnormal trading events**.

Event Study Analysis

An event study was conducted to examine abnormal returns around these suspicious events. The event window spans from **10 trading days before to 10 trading days after each event**:

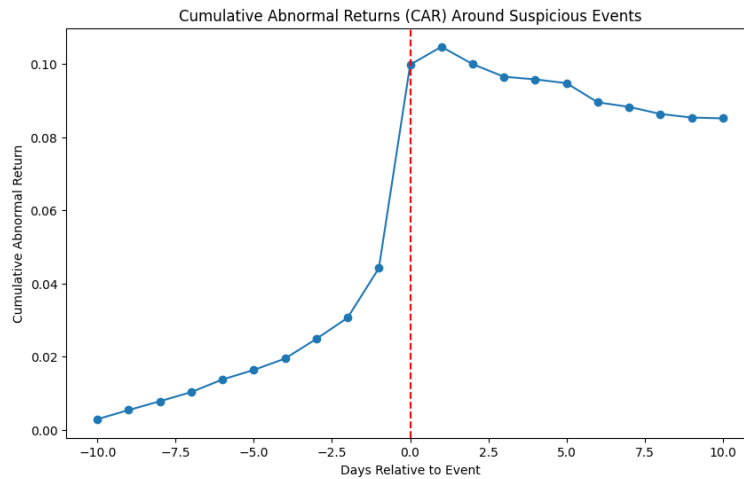
$$t \in [-10, +10]$$

where ($t = 0$) represents the abnormal trading day.

Two key measures were calculated:

- **Average Abnormal Return (AAR)**
- **Cumulative Abnormal Return (CAR)**

Figure 1. Cumulative Abnormal Returns Around Suspicious Trading Events



The cumulative abnormal return in the five days preceding the event is:

$$CAR_{(t_1, t_2)} = \sum_{t=t_1}^{t_2} A R_{i,t}$$

Statistical test results:

- **t-statistic = 9.15**
- **p-value < 0.001**

This result indicates that stock prices tend to increase significantly before abnormal trading events.

The cumulative abnormal return after the event is:

Pre-event cumulative abnormal return

$$CAR(-5, -1)$$

Post-event cumulative abnormal return

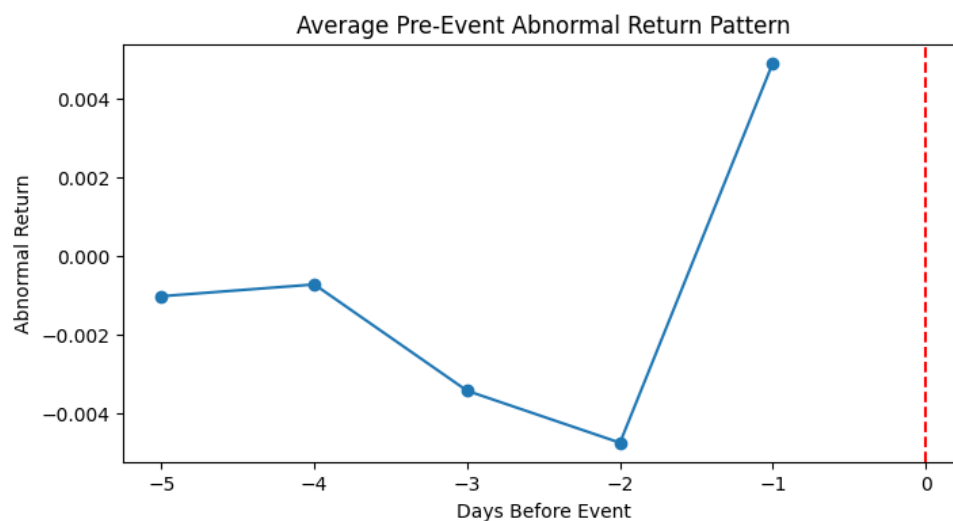
$$CAR(+1, +5)$$

Statistical test results:

- **t-statistic $\approx$ -1.64**
- **p-value $\approx$ 0.10**

This suggests a mild but statistically weak price reversal following the event.

Figure 2. Average Abnormal Returns Around Events



The average abnormal return plot shows a clear spike at the event day and positive abnormal returns in the days leading up to the event.

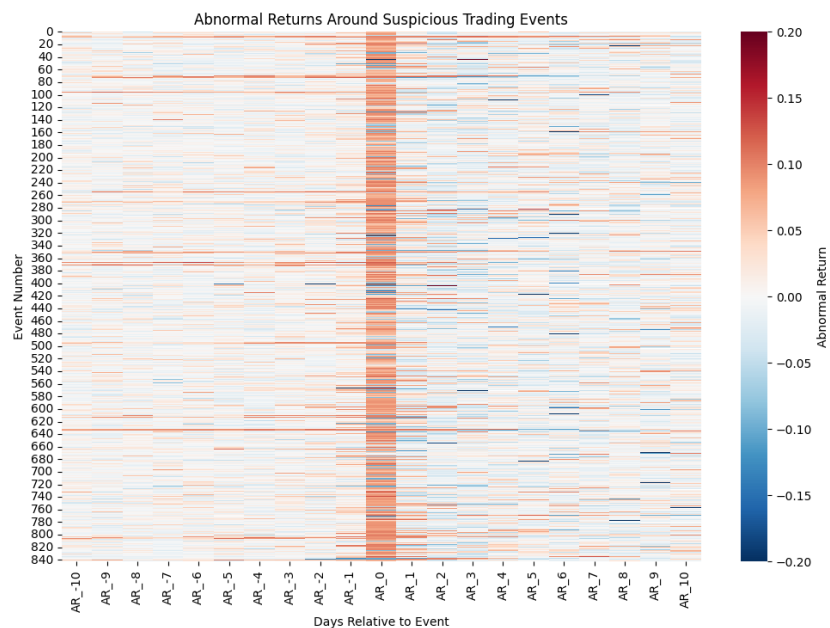
Table 2. Event Study Results

Event Window Mean CAR t-statistic		
CAR(-5,-1)	3.04%	9.15
CAR(+1,+5)	-0.50%	-1.64

Visual Evidence of Trading Behavior

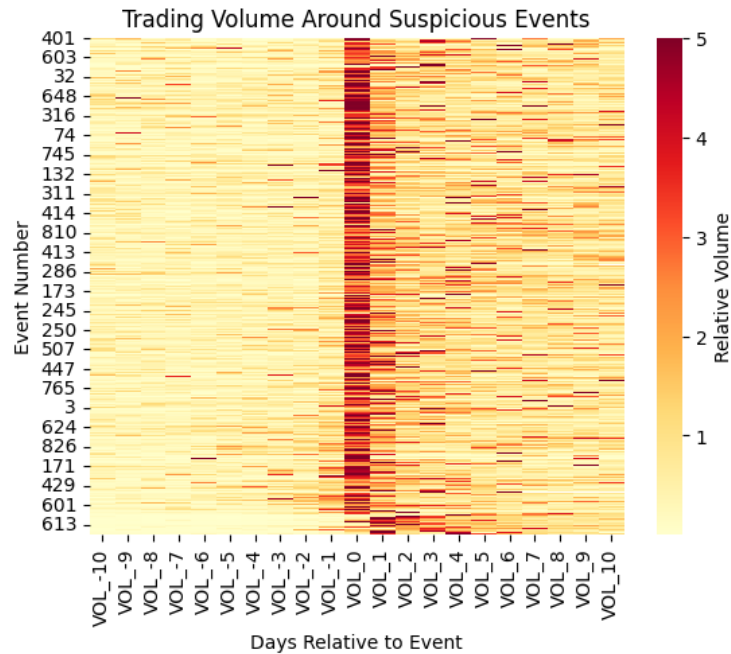
Visual analysis provides additional insight into trading dynamics around abnormal events.

Figure 3. Heatmap of Abnormal Returns Around Events



The heatmap displays abnormal returns for all detected events across the event window. Many events exhibit positive abnormal returns prior to the event day, suggesting systematic price buildup before abnormal trading activity occurs.

Figure 4. Trading Volume Around Suspicious Events



The volume heatmap reveals a large increase in trading activity on the event day, accompanied by elevated trading levels in surrounding days. This pattern indicates intensified market participation during abnormal trading events.

Interpretation

The combined results reveal a consistent pattern:

- Stock prices tend to increase before abnormal trading events.
- Event days are associated with significant price and volume spikes.
- Prices partially revert following the event.

This pattern may reflect several possible mechanisms, including information leakage, informed trading behavior, or speculative trading episodes within the market.

Next Steps

The next phase of this research will focus on two complementary analyses.

Insider Trading Detection

This analysis will examine whether abnormal price and volume movements prior to events are consistent with informed trading behavior.

Pump-and-Dump Detection

This analysis will identify cases where sharp price increases are followed by rapid price declines, potentially indicating coordinated market manipulation.

Additionally, suspicious events will be cross-referenced with corporate announcements such as:

- dividend announcements
- bonus share issuances
- rights offerings
- book closures

This step will help distinguish legitimate market reactions from potentially abnormal trading activity.

Conclusion

This study provides preliminary evidence of systematic abnormal trading behavior in the Nepal Stock Exchange (NEPSE). Using an event study framework, the analysis identifies 843 abnormal trading events characterized by simultaneous spikes in price and trading volume. The results reveal statistically significant cumulative abnormal returns of approximately 3.04% in the five trading days preceding these events, suggesting that stock prices tend to increase prior to abnormal trading activity. Visual analysis of abnormal return and volume heatmaps further indicates that many events exhibit price and trading volume buildup before the event day.

These findings are consistent with patterns that may reflect information leakage, informed trading, or speculative trading activity in the market. Future work will extend this analysis by distinguishing between potential insider trading and pump-and-dump behavior, as well as cross-referencing detected events with corporate announcements such as dividends, bonus shares, rights issues, and book closures.

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